

CONTROL SYSTEM DESIGN FOR MITIGATING PEDESTRIAN INDUCED LATERAL VIBRATIONS IN FOOTBRIDGES

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Project Topic: [Design of a Control System for Active Footbridge Stabilisation]

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Abstract: This paper presents the design and evaluation of a control system aimed at mitigating pedestrian induced lateral vibrations in footbridges, which have been linked to serviceability failures and in extreme cases collapse. The study focuses exclusively on the lateral pedestrian-structure interaction and employs a simulation-based solution using MATLAB and Simulink. Three control laws are proposed. As a proof of concept, the first two are tested on a simplified single degree of freedom footbridge model. The approach is then extended to a more realistic multi degree of freedom representation of the bridge, where multiple actuators are coordinated using a total sum penalty Linear Quadratic Regulator control law. The performance of the multi actuator controller is assessed against international serviceability limits and its energy demands are examined to evaluate feasibility. The results demonstrate that the proposed controller effectively mitigates footbridge vibrations across a wide frequency range while maintaining system stability. The table provided in appendix A provides navigation on the addressed graduate attributed for this report.

Keywords: Lateral pedestrian forces, Linear Quadratic Regulator, MATLAB, Serviceability limits

1. INTRODUCTION

Footbridges once simple structures designed to allow passage across gaps in terrain, have evolved into more advanced structures capable of carrying heavier loads and serving diverse functions beyond pedestrian use. Modern engineering trends favour lighter, more slender and more flexible designs but these innovations also introduce new challenges, in particular a high susceptibility to excessive vibrations [1]. Uncontrolled vibrations can compromise safety by accelerating material fatigue and reduce structural integrity. Even in less severe cases, they result in serviceability problems such as cracks, loose fittings and material wear which increase maintenance costs and undermine public confidence in the design. In extreme cases, vibrations may result in catastrophic failure and loss of life, as seen in the collapse of the Ponte das Barcas, which drowned approximately 4000 people and the Yarmouth suspension bridge collapse, which killed 79. Extensive research over centuries has however identified several types of vibrations in footbridges that contribute to such collapses, namely vertical [2], torsional [3], longitudinal [4] and lateral [1] vibrations. While vertical and torsional vibrations are relatively well understood and addressed through established design codes and mitigation strategies, lateral vibrations remain difficult to predict and model accurately for analysis. Primarily due to the phenomenon of pedestrian synchronisation where collective pedestrian movement amplifies structural motion. The aim of this report is to investigate pedestrian induced lateral vibrations in footbridges and evaluate mitigation strategies for improving serviceability and safety. The scope of the analysis is limited to lateral loading with vertical, longitudinal and torsional coupling effects considered outside the focus of this report.

The phenomenon of lateral synchronized excitation arises in footbridges with a natural frequency below approximately 1.3 Hz [5], particularly when the crowd density exceeds $0.2 \text{ walkers m}^{-2}$ [6]. Under these conditions, pedestrian-structure interaction produces noticeable lateral vibrations. When vibration levels surpass typical comfort thresholds, pedestrians subconsciously adjust their walking pace and rhythm to align with the bridge's lateral motion. This collective synchronization amplifies the total lateral force acting on the structure, effectively creating a resonant input that continuously feeds energy into the system. As a result, the bridge vibrations can escalate and in extreme cases approach levels that threaten structural stability.

To maintain safety and comfort, vibration control systems are required to mitigate footbridge instability due to crowd loading. Passive solutions such as tuned mass dampers (TMDs) can reduce structural response by oscillating out of phase with the bridge thus absorbing and dissipating kinetic energy. However, their performance degrades significantly when used to control low amplitude human induced vibrations [7] resulting in a short term solution. In contrast active systems in particular the active tuned mass damper (ATMD) can adapt in real time using sensors, controllers and actuators to counteract vibrations. This adaptability makes ATMDs a promising solution to mitigating pedestrian induced lateral sway. This report therefore will focus on the application of ATMDs on footbridges subject only to pedestrian induced lateral excitation. The initial analysis concentrates on the midspan of the footbridge, where lateral displacements and pedestrian loading are greatest [1], [8] with the final model focusing on the analysis of a proposed discretised footbridge model. The first footbridge model is modelled as a single degree of freedom (SDOF) system subject to harmonic pedestrian excitation in the range 0.7 – 1.3 Hz. Structural behaviour is assumed to be linear and elastic, damping is taken as viscous

and proportional and pedestrian interaction is modelled through synchronisation effects rather than individual gait variability. These assumptions provide a simplified yet representative framework for evaluating the bridge and pedestrian interaction while demonstrating the effectiveness of active control strategies. The report is structured as follows, section 2 presents the theoretical background and governing equations for the single degree of freedom (SDOF) system and analysis of the applied vibration mitigation strategies, section 3 details the formulation of the N-DOF footbridge model and analysis of the results obtained from MATLAB and Simulink. Section 4 provides a critical serviceability analysis, energy considerations of the proposed control system and design evaluation in regards to proposed design specifications. Section 5 focuses on the social, environmental and economic impact of the proposed control system while section 6 provides the sustainability analysis and section 7 concludes the report.

2. ACTIVE VIBRATION CONTROL

2.1 Footbridge Midspan Dynamics

The footbridge midspan is limited to only 6m of the 30 m length and is modelled as a SDOF system to analyse the dynamic response to pedestrian induced vibrations. The SDOF Simulink model is shown in appendix E and this SDOF model is defined by the following equation:

$$M\ddot{x} + B\dot{x} + Kx = F_p \quad (1)$$

where M is the modal mass of the footbridge, B is the modal damping of the footbridge, K is the modal stiffness of the footbridge and F_p is the pedestrian lateral force. The induced lateral force as proposed by HIVOSS guidelines [9] is modelled by the following equation:

$$F_p(t) = G \cos(2\pi ft) * n^1 * \psi \quad (2)$$

$$n^1 = \frac{10.8\sqrt{\zeta * n}}{s} \quad (3)$$

The harmonic load factor due to a single pedestrian s expressed as $G \cos(2\pi ft)$, where $G = 35N$. The equivalent number of pedestrians per unit area contributing to the lateral force is represented by n^1 , and the reduction coefficient of $\psi = 0.8$ is applied to account for interaction effects. In equation (3), the damping factor is taken as $\zeta = 5\%$, the total number of pedestrians is denoted by n , and the occupied area of the dense crowd in the simulations fixed at $S = 18m^2$. The modal values that were used for the simulation of the dynamic response of the footbridge are a modal mass of 294.375 kg , modal stiffness of $11621.459 \text{ Nm}^{-1}$ a modal damping of $1849.612 \text{ Nsm}^{-1}$ ensuring that the natural frequency of the bridge is below the 1.3 Hz limit for lateral excitation as shown by the equation:

$$f_n = \frac{1}{2\pi} * \sqrt{\frac{K}{M}} = \frac{1}{2\pi} * \sqrt{\frac{11621.459}{294.375}} = 0.999 \text{ Hz} \quad (4)$$

The response of the footbridge midspan to a lateral force at the natural frequency to a 200-person crowd is presented below.

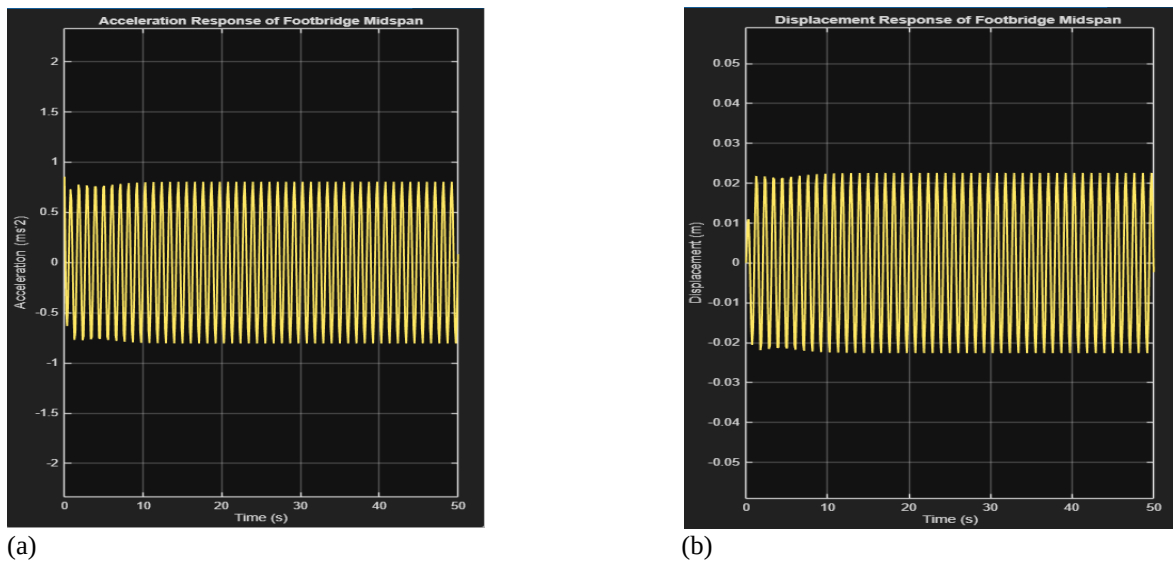


Figure 1: (a) Acceleration response to natural frequency (b) Displacement response to the natural frequency

Figure 1a shows the acceleration response of the midspan to the natural frequency showing a peak value of 0.805 ms^{-2} and figure 1b shows the displacement produced at this frequency being a peak value of 22.6 mm. These values highlight the uncontrolled natural response providing a baseline against which the effectiveness of the proposed control strategies can be evaluated.

2.2 Actuator & Sensor

The ATMD is mounted under the midspan of the bridge deck because at this location modal displacements are largest thus maximising its effectiveness. The SDOF configuration for the mounted ATMD is shown in the figure below.

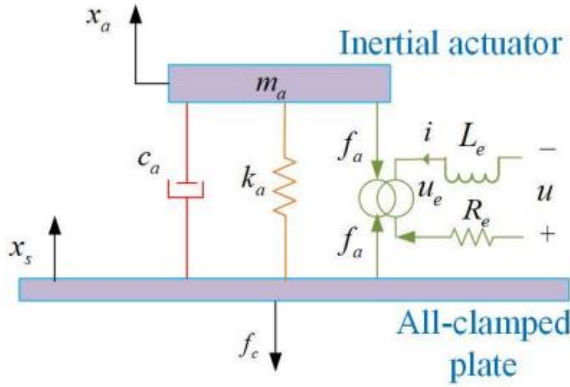


Figure 2 : Active tuned mass damper mounted to bridge deck [10] .

The ATMD is often realised as an electromagnetic inertia actuator, operating on the principle that an input voltage applied to the device is converted into a mechanical control force. The actuator consists of a coil of effective length, l , laced within a constant magnetic flux density B generated by a permanent magnet. When the input voltage u shown in figure 2 is applied across the coil terminals the electrical circuit dynamics are described by the equation:

$$u = Bl(\dot{x}_a - \dot{x}_s) + R_e i + L_e \frac{di}{dt} \quad (5)$$

where i is the current in the coil, R_e and L_e are the resistance and inductance of the actuator circuit and the term $Bl(\dot{x}_a - \dot{x}_s)$ represents the back electromotive force induced by the relative velocity between the actuator mass and the structure. The current i flowing through the coil produces an actuation force governed by Lorentz law:

$$f_a = Bli \quad (6)$$

This force acts directly on the moving mass of the actuator, which can be modelled mechanically as a SDOF oscillator with a mass m , damping c and stiffness k with its governing equation between the structure and actuator:

$$m_a \ddot{x}_a + c_a(\dot{x}_s - \dot{x}_a) + k_a(x_s - x_a) \quad (7)$$

The contact force transmitted from the actuator to the structure is governed by the equation:

$$f_c = f_a + c_a(\dot{x}_s - \dot{x}_a) + k_a(x_s - x_a) \quad (8)$$

The passive components of the ATMD are optimally tuned to the specific natural frequencies of the footbridge using Dan Hartog optimisation equations as follows:

$$m_a = \mu * m_s = 0.04 * 294.375 = 11.775 \text{ kg} \quad (9)$$

$$\delta_a = \frac{1}{1+\mu} = 0.9615 \quad (10)$$

$$\zeta_a = \sqrt{\frac{3\mu}{8(1+\mu)}} = 0.12 \quad (11)$$

The mechanical parameters of the ATMD can be determined by the equations:

$$c_a = 4 * m_a * \pi * \delta_a * f_s * \zeta_a \quad (12)$$

$$k_a = m_a * (2 * \pi * \delta_a * f_s)^2 \quad (13)$$

where c_a and k_a are the damping and stiffness of the ATMD, m_a is the mass of the ATMD, δ_a tuning frequency ratio between the structure and the ATMD frequency, f_s is the natural frequency of the structure and ζ_a is the damping ratio of the ATMD. To enable feedback control of the ATMD, the structural response must be measured in real time. In this study a piezoelectric accelerometer is adopted as the primary sensing device, to be distributed along the length of the bridge for measurements of acceleration at each point mass. Piezoelectric sensors operate on the principle that mechanical stress applied to a piezoelectric material generate a proportional electrical output typically in the form of a voltage. This makes them well suited for vibration monitoring due to their high sensitivity and operation within a wide frequency range. For the proposed model the accelerometer is governed by the relation:

$$E = gt \frac{F}{A} \quad (14)$$

where g is the material voltage sensitivity, t is the piezoelectric layer thickness, A is the effective sensing area and F is the applied inertial force due to the measured acceleration. The inertia force acting on the sensing element is expressed by the equation $F = m_{eq}a$ where m_{eq} denotes the equivalent mass of the accelerometer. In the proposed model, the accelerometer output voltage, E , is directly processed by the control system and subsequently used as the input driving signal for the controller to produce the actuating force. In this way, the measured bridge acceleration is transformed into an electrical command that governs the actuator's response, ensuring that the ATMD force is continuously adapted to the real-time structural motion.

The parameters governing the ATMD at the natural frequency of the SDOF model, and the accelerometer are expressed in table 1 and table 2 below.

Table 1: ATMD parameters

m_a (kg)	c_a (Nms ⁻¹)	k_a (Nm ⁻¹)	Bl (m)	L_e (H)	R_e (Ω)
12.775	16.4182	429.753	30.4	0.002	7.5

Table 2: Accelerometer parameters

Voltage Sensitivity (VmN ⁻¹)	Thickness (m)	Area (m ²)
$25 * 10^{-3}$	$2 * 10^{-3}$	$25 * 10^{-6}$

2.3 Single Degree of Freedom PID Controller

The desired control force to be applied by the ATMD is determined by calculating an equivalent active interaction force. In this system the active force is generated through a PID controller, chosen for its effectiveness and ease of implementation. The PID controller operates on the basis of a feedback error, denoted as $e(t)$. For negative feedback system the error is defined as the difference between the desired reference signal $r(t)$ and the measured signal $y(t)$ from the accelerometer such that:

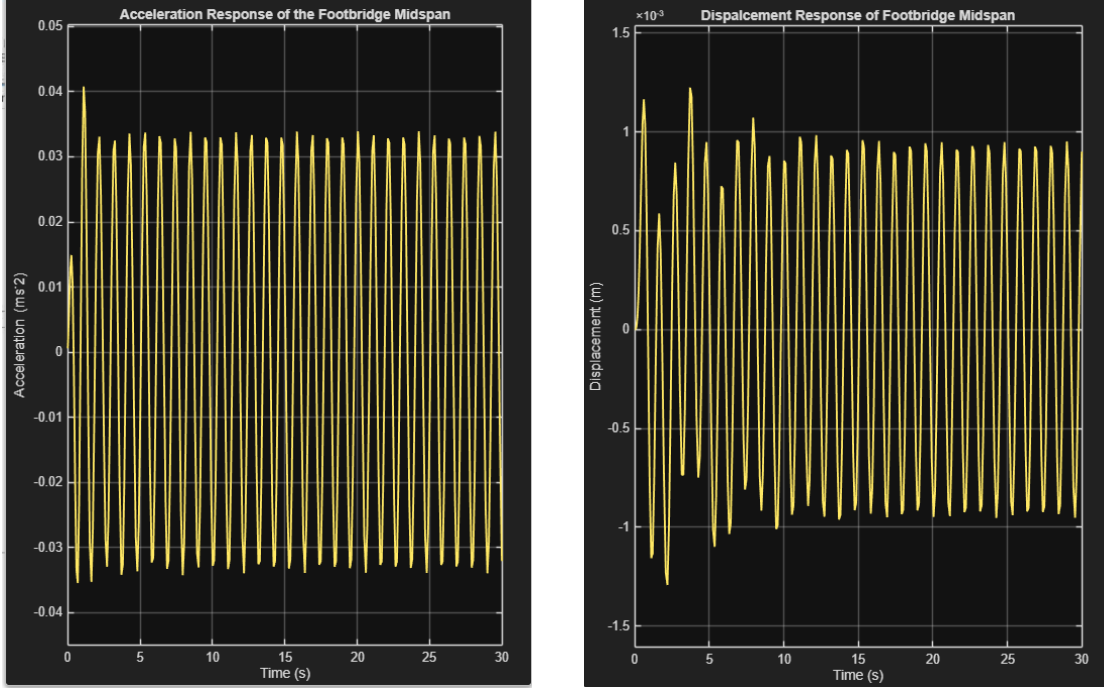
$$e(t) = r(t) - y(t) \quad (15)$$

In this control system the desired reference state will correspond to a zero-acceleration equilibrium position, hence $r(t) = 0$. The role of the controller is to minimise this error by adjusting the control input $u(t)$ which represents the voltage signal applied to the ATMD circuit to generate the actuating contact force. Following the standard formulation of the PID controller the control voltage is then expressed as:

$$u(t) = K_p e(t) + K_i \int_0^t e(t) dt + K_d \frac{de(t)}{dt} \quad (16)$$

where K_p , K_i and K_d are the proportional, integral and derivative gains respectively. Manual tuning was used to determine suitable values for the PID controller implemented on the SDOF bridge midspan. The values that provided good control performance were determined to be a proportional gain of 20, integral gain of 2 and a derivative gain of 1. The deck response to the lateral force produced by a 200-person crowd is shown in figure 3, showing the displacement and

acceleration response. In figure 3b, the maximum displacement is shown to be 1.22 mm, while figure 3a shows the acceleration response produced by the lateral force, peaking at 0.0408 ms^{-2} . A comparison with the results obtained in figure 1 shows that the PID effectively reduces the displacement by 94.6 % while the acceleration is reduced by 94.9 %.



(a)

(b)

Figure 3: (a) Acceleration response of footbridge midspan (b) Displacement response of footbridge midspan

The resulting SDOF model for the PID controller can be visualised in appendix F. In addition to the PID controlled SDOF model, a SDOF model with a Linear Quadratic Regulator (LQR) control law was also developed. This provides a frame work of comparison in performance between the two control schemes in an effort to extend the complexity of the bridge to a more realistic model that incorporates multiple modes of vibration reflecting real world application.

2.4 Single Degree of Freedom LQR Controller

The desired control force can also be provided through an optimal control approach. In this framework the control input is derived by solving an optimisation problem that minimises a cost function balancing system performance and controller effort. The LQR controller operates on the basis of a state feedback control law, where the error is not calculated in the form of a scalar deviation $e(t)$, but rather through the systems state vector denoted as $x(t)$. The cost function minimised by the LQR controller is of the form:

$$J = \int_0^\infty (x(t)^T Q x(t) + u(t)^T R u(t)) dt \quad (17)$$

where Q is a positive semi-definite matrix that penalises the system deviations in the state equation thereby influencing how strongly the controller suppresses structural vibration. The weighting matrix R is a positive definite scalar or matrix that penalises excessive control effort to prevent excessive actuator input. Through the tuning of the Q and R matrix a balance can be struck between vibration suppression and energy consumption of the actuator. The optimal control force is then expressed as:

$$u(t) = -Kx(t) \quad (18)$$

where K is the optimal gain matrix obtained by solving the algebraic Riccati equation associated with the system dynamics. This feedback law ensures that the control input is directly proportional to a weighted combination of displacement and velocity, with the weights optimally determined by the LQR design. Unlike the PID controller which requires tuning of the K_p, K_i and K_d parameters the LQR controller systematically generates the feedback gain matrix K from the solution of the optimisation problem leading to guaranteed stability and optimality for the given weighting

parameters. The state matrices used for the formulation of the optimal controller are produced by using the matrix form of the SDOF equations presented below while the Simulink format is visualised in appendix G.

$$[M] \ddot{x} + [C] \dot{x} + K x = [F_p] + [F_a] \quad (19)$$

where M is the mass matrix, K the stiffness matrix, C the damping matrix, Fp the lateral force matrix and Fa the actuating force produced by the ATMD. These matrices can be better expanded as shown below:

$$\begin{bmatrix} m_s & 0 \\ 0 & m_a \end{bmatrix} \begin{bmatrix} \ddot{x}_s \\ \ddot{x}_a \end{bmatrix} + \begin{bmatrix} c_s + c_a & -c_a \\ -c_a & c_a \end{bmatrix} \begin{bmatrix} \dot{x}_s \\ \dot{x}_a \end{bmatrix} + \begin{bmatrix} k_s + k_a & -k_a \\ -k_a & k_a \end{bmatrix} \begin{bmatrix} x_s \\ x_a \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} F_p + \begin{bmatrix} -1 \\ 1 \end{bmatrix} F_a \quad (20)$$

The state space transformation for the dynamic matrices is presented below.

$$\dot{x} = [A] x(t) + [B_u] u(t) + [B_d] d(t) \quad (21)$$

$$y = [E] x(t) \quad (22)$$

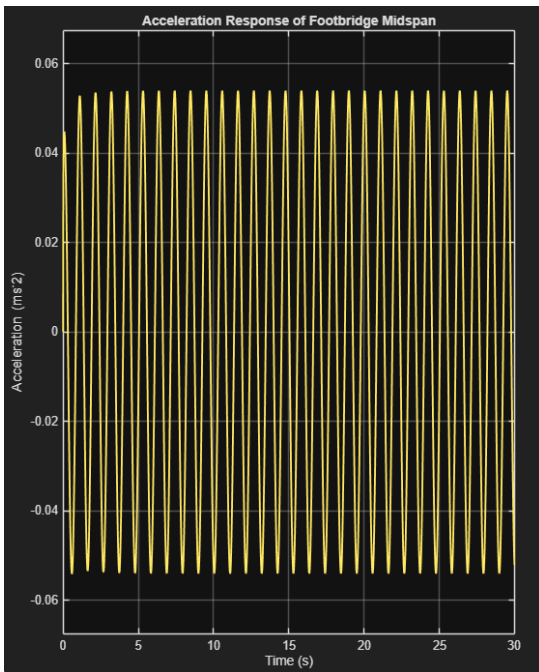
$$[A] = \begin{bmatrix} 0 & I \\ -[M]^{-1}[K] & -[M]^{-1}[C] \end{bmatrix} \quad (23)$$

$$[B_d] = \begin{bmatrix} 0 \\ -[M]^{-1}[F_p] \end{bmatrix} \quad (24)$$

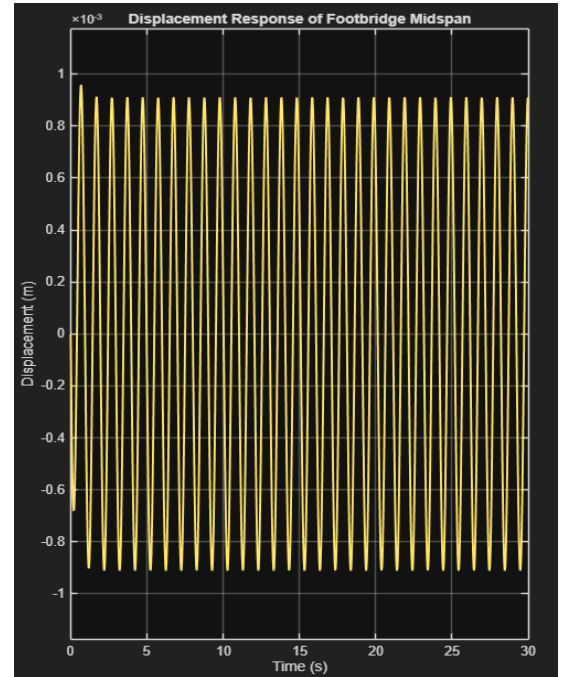
$$[B_u] = \begin{bmatrix} 0 \\ -[M]^{-1}[F_a] \end{bmatrix} \quad (25)$$

$$[E] = -[E_a][M]^{-1}[K] \quad [M]^{-1}[C] \quad (26)$$

where [A] is the state matrix, [Bd] the input matrix for the lateral force, [Bu] is the input matrix for the control force, [E] output matrix and [Ea] is the acceleration matrix which indicates the acceleration of different elements in the dynamic equations. The state vector defines the displacement and velocities of the footbridge and the ATMD presented in the form $\{x(t)\} = \{x_s \ x_a \ \dot{x}_s \ \dot{x}_a\}$. The lqr function in MATLAB was used to find the optimal control gain K. The dynamic response of the optimal controller for a 200-person crowd is shown below.



(a)



(b)

Figure 4: (a) Acceleration response of footbridge midspan (b) Displacement response of footbridge midspan

The dynamic response of the bridge midspan under lateral force in figure 4a shows a peak acceleration of 0.053 ms^{-2} , representing a 93.4 % reduction compared to the natural response in figure 1a. The corresponding displacement in figure 4b reaches a maximum of 1 mm which is a 95.7 % improvement relative to the uncontrolled response shown in figure 1b. When compared with the conventional PID controller, which achieves a 94.9 % reduction in acceleration and a 94.6 % reduction in displacement the proposed controller demonstrates comparable suppression performance while maintaining robustness. This highlights the effectiveness of the ATMD as a vibration mitigation system. While these results confirm the effectiveness of the proposed controllers, they are derived from a SDOF model which provides only a simplified representation of the bridge by focusing on a single fundamental vibration mode. Such a model is useful for preliminary analysis and controller design, but it presents significant limitations being that it cannot capture higher order modes excited by pedestrian loading nor can it account for mode coupling or the spatial distribution of pedestrian forces. These oversimplifications restrict its ability to represent the full dynamic behaviour of a real footbridge. To address this the next section develops a more detailed discretised model using coupled masses, springs and dampers. This formulation not only captures the complex dynamics more accurately but also allows for the application of an optimal control law expressed naturally in state space form.

2.5 Footbridge Deck Modelling

The bridge deck is modelled as a distributed mass system where each point mass is connected to adjacent masses through springs and dampers representing lateral stiffness and damping characteristics. A key challenge in modelling lateral motions is that spring forces depend on angular displacement, introducing geometric nonlinearities. However, if the structural elements are highly pretensioned from their relaxed length without distortion [11], the lateral vibration frequencies become nearly identical to those of longitudinal vibrations, suggesting that linear analysis methods remain valid. This observation motivated the assumption that the structural elements are highly pretensioned relative to anticipated lateral displacements. Under this assumption, small lateral displacements result in spring forces that are approximately linear with respect to displacements as shown with the equations below. This linearisation is physically justified for cable stayed bridges or suspension footbridges where the initial tension far exceeds the forces from pedestrian induced lateral motion. This linearisation also allows for the implementation of the linear optimal control law. The linearisation is provided below.

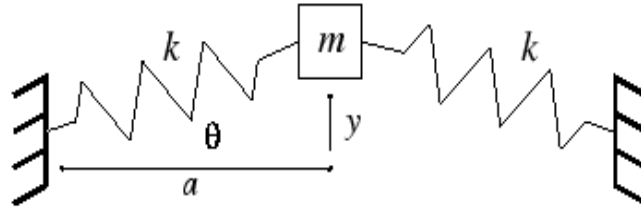


Figure 5: Birds-eye view of a single mass with two springs system for lateral motion [11].

Considering the spring mass configuration provided in figure 5, at a natural length a and pre-stretched to a length Δy from the equilibrium position the length of each spring can be provided by the equation:

$$length = \sqrt{a^2 + \Delta y^2} \quad (27)$$

which can be visualised by the hypotenuse in figure 5. The extension of each spring can then be modelled as:

$$extension = \sqrt{a^2 + \Delta y^2} - a \quad (28)$$

As a result of the configuration, a restoring force is produced by each spring in the vertical direction to pull the mass back to the equilibrium position. The magnitude of the net restoring force on the mass is then:

$$|F_{restoring}| = k(\sqrt{a^2 + \Delta y^2} - a) \sin \theta \quad (29)$$

$$|F_{net \ force}| = 2k(\sqrt{a^2 + \Delta y^2} - a) \sin \theta = 2k(a + \Delta y - a) \sin \theta \quad (30)$$

$$|F_{net}| = 2k \Delta y \sin \theta \quad (31)$$

by substituting $\sin \theta$ which is provided by geometry as:

$$\sin \theta = \frac{\Delta y}{\sqrt{a^2 + \Delta y^2}} \quad (32)$$

the resulting net force equation becomes:

$$|F_{net}| = 2k \frac{\Delta y^2}{\sqrt{(a^2 + \Delta y^2)}} \quad (33)$$

The assumption is that the spring is highly pretension, meaning Δy is much larger than the natural length making it a very small number and by approximation $a \approx 0$, the resulting net force equation becomes:

$$|F_{net}| = 2k \frac{\Delta y^2}{\sqrt{\Delta y^2}} = 2k\Delta y \quad (34)$$

This shows that under high pretension the spring behaves as if it were linear in the transverse direction while remaining physically realistic for cable supported bridges. It is to be noted that this same format is extended for a mass connected to two dampers to become $|F_{net}| = 2c\Delta\dot{y}$, as the damping force is proportional to velocity. The proposed single mass model is then extended to N-DOF segments to represent the footbridge's coupled nature, this is represented by the matrix equations:

$$[M] \ddot{y} + [C] \dot{y} + [K] y = [F_p] + [F_a] \quad (35)$$

where $[M]$ is the diagonal mass matrix for point masses M_1 to M_N , $[K]$ is a tri-diagonal stiffness matrix for the spring's constants K_1 up to K_{N+1} , with the diagonal elements the sum of the two spring constant values connected to each point mass. $[C]$ a tri-diagonal damping matrix for the dampers C_1 up to C_{N+1} with the diagonal elements being the sum of the damping coefficients of the dampers connected to each point mass. The disturbance matrix F_p , maps the lateral force to different point masses corresponding to the shape of the excited mode of vibration on the footbridge to test the robustness of the control law. The actuator matrix F_a contains the locations of the distributed ATMD's coupled to different point masses. The matrices for the N-DOF model are then transformed into the state space format using the equations (21) – (26), to allow for integration with an LQR controller. For the N-DOF model different vibrational modes are excited by constructing input vectors consistent with the corresponding mode shape. For each mode shape, the normalised mode shape is thresholded so that only the point masses with significant participation receive the lateral force. This excitation pattern is implemented through the input matrix B_d (21), ensuring that applied forces reproduce the spatial distribution of the selected mode. The parameters that define the bridge model are detailed in following table.

Table 3: Footbridge deck parameters

Point mass (kg)	Support stiffness (Nm)	Deck stiffness (Nm)	Support damping (Nsm ⁻¹)	Internal damper (Nsm ⁻¹)
294.375	58107.295	11621.459	9248.06	1849.612

The support stiffness and damping values in table 3 are five times bigger than the deck values as the anchor points of the bridge do not move significantly when the bridge is laterally loaded.

2.6 N-DOF LQR Controller

The actuators are attached to the 3rd, 7th, and 12th point masses to maximize control authority across the dominant lateral modes. Placement at the 7th point mass (midspan) coincides with the antinodes of odd numbered modes, ensuring efficient suppression of their peak responses. The additional actuators at the 3rd and 12th point masses target antinodes of even modes and higher order responses, while also avoiding nodal locations where actuator influence would be minimal. Together, this configuration provides broad modal coverage, distributing the required control force to reduce per actuator demand, and achieving an effective balance between vibration mitigation and implementation practicality. While the standard cost function provided by equation (17) is effective for the control of one actuator it proves ineffective for multi actuator control, as it results in the generation of a net force that may excite the bridge or cancel out due to a lack of synchronicity. To address this an additional penalty term is introduced below to target the sum of all actuator forces, discouraging the actuators from working against each other. In the standard cost function (17), the term $u^T R u$ penalises control effort. For a diagonal R , this expands to:

$$u^T R u = u^T \begin{bmatrix} r_1 & 0 & 0 \\ 0 & r_2 & 0 \\ 0 & 0 & r_3 \end{bmatrix} u = r_1 u_1^2 + r_2 u_2^2 + r_3 u_3^2 \quad (36)$$

which only penalises the squared magnitude of each actuators force individually. The cost function that penalises the sum force is proposed below.

$$u_{sum} = u_1 + u_2 + u_3 + u_4 + \dots + u_n = 1^T u \quad (37)$$

where 1 is a column vector of ones. In penalising the squared magnitude of the total control effort, the cost function becomes:

$$J = \int_0^\infty (x^T Q x + u^T R u + \lambda_{sum} (1^T u)^2) dt \quad (38)$$

where λ_{sum} is the penalty on the sum of all actuator forces that discourages situations where the controller cannot counteract the disturbance force efficiently. The tuning of the Q, R and λ_{sum} penalties was carried out iteratively in a structured sequence. Initial values emphasised state penalties Q to reduce displacement and acceleration responses, followed by adjustments of R to balance performance with actuator effort, and final refinements of λ_{sum} to distribute control across all vibration modes. This systematic process ensured convergence to a control law that achieved vibration suppression while remaining within practical power limits. The values determined by this iterative method are shown in table 4 below:

Table 4: LQR weight penalties

Deck displacement (Q)	Deck velocity (Q)	ATMD displacement (Q)	ATMD velocity (Q)	R	λ_{sum}
0.0001	1000	0.001	0.001	1000	0.0001

The effectiveness of the proposed control strategy for the realistic footbridge model was judged against the following design specifications.

Table 5: Design specifications

Specification	Description
Lateral acceleration	The system shall reduce lateral accelerations of the bridge deck to below the unacceptable serviceability comfort limit specified by international guidelines ($\leq 0.80 \text{ ms}^{-2}$ under dense pedestrian loading).
Stability	The closed loop system shall remain stable for all expected pedestrian loading scenarios.
Actuation voltage	Actuators shall operate within a maximum input voltage range of $\pm 50 \text{ V}$.
Actuation current	Actuation shall operate within a maximum rated current of 10A.
Power consumption	Average power consumption per ATMD shall not exceed 500 W, with sufficient provision for transient peak demand.

3. N-DOF SIMULATION

The analysis of the control law for the N-DOF model, requires the identification of the natural frequencies that can be excited on the footbridge due to pedestrian loading. Using the mass and stiffness matrices through modal analysis the natural frequencies of the footbridge can be identified by solving the following eigenvalue problem:

$$(K - \omega^2 M)\phi = 0 \quad (39)$$

$$\phi = \frac{\sin(2x)}{L} \quad (40)$$

where ω is the natural frequency values, ϕ is the corresponding mode shape and L is the length of the bridge deck. The eigenvalue problem was solved using MATLAB and the full spectrum of natural frequencies and corresponding shapes are provided in the table below.

Table 6: Computed natural frequencies of the N-DOF bridge model.

Mode	Frequency	Damping ratio	Critical Pedestrian's	Description
1	0.2527	1.5879	10	First lateral mode
2	0.5016	3.1519	39	Second lateral mode
3	0.7430	4.6683	86	Third lateral mode
4	0.9731	6.1141	147	Fourth lateral mode
5	1.1884	7.4667	219	Fifth lateral mode
6	1.3854	8.7044	297	Sixth lateral mode
7	1.5607	9.8064	377	Seventh lateral mode
8	1.7112	10.7520	454	Eighth lateral mode
9	1.8336	11.5209	520	Nineth lateral mode
10	1.9246	12.0924	574	Tenth lateral mode
11	1.9809	12.4463	608	Eleventh lateral mode
12	2.5000	15.7080	968	Twelveth lateral mode
13	2.5030	15.7085	969	Thirteenth lateral mode

The corresponding mode shapes provided in table 6 illustrate how displacement patterns develop along the span of the bridge. The first mode is dominated by a single half-sine deflection profile, the second mode exhibits one internal node and the third two internal nodes producing a pattern of N internal nodes equal to the mode excited, a visualisation of the first few vibration modes are provided in appendix B. Excessive lateral acceleration only occurs in structures with natural a natural frequency between the range 0.4 – 1.3 Hz [12], in cases when the number of pedestrians walk at a natural frequency of the bridge and are above a critical number which can be estimated by the following equation:

$$N_{critical} = \frac{8\pi\zeta\omega m}{k} \quad (41)$$

where m is the point mass, k = 300 is the lateral walking force proportionality factor [1], ζ and ω are the damping ratio and frequency of the lateral mode of interest. Since the lowest value of lateral excitation is 0.4 Hz this paper will not analyse the first lateral mode within the main body, the uncontrolled and controlled responses to the controller are provided in appendix C1 – C7. The main body will only address the effect of the control law for the second to sixth lateral modes. To ensure the controllability and stability margins of the system, a closed loop analysis was done in MATLAB using the A matrix, B matrix and the LQR feedback gain matrix to interpret the eigen values of the closed loop system matrix using the following equations:

$$A_{CL} = [A] - [B] * [K] \quad (42)$$

$$\det(A_{CL} - \lambda I) = 0 \quad (43)$$

where λ are the corresponding eigenvalue for the system dynamics. By ensuring that all poles obtained from the eigenvalue have negative real parts the stability and controllability of the model is ensured. The obtained eigenvalues ensuring the systems stability and controllability are provided in appendix D.

3.1 2nd Lateral Vibration Mode

The response of the footbridge to the second lateral frequency with the calculated critical number of pedestrians provided in table 6 is shown in the following figures.

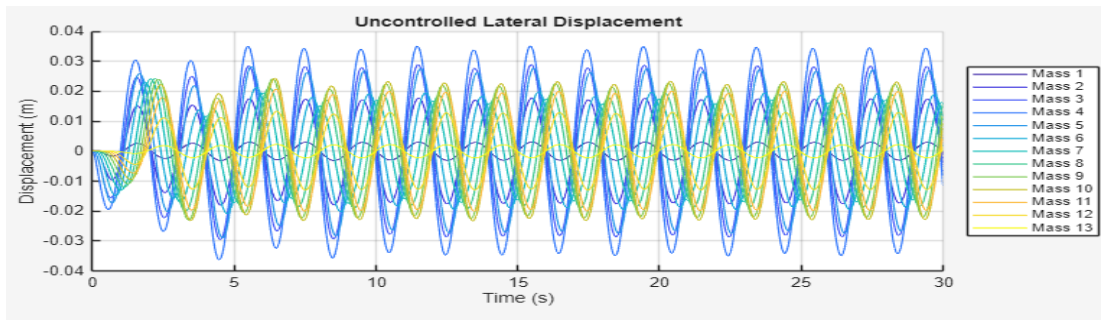


Figure 6: Uncontrolled transverse displacement response to pedestrian input

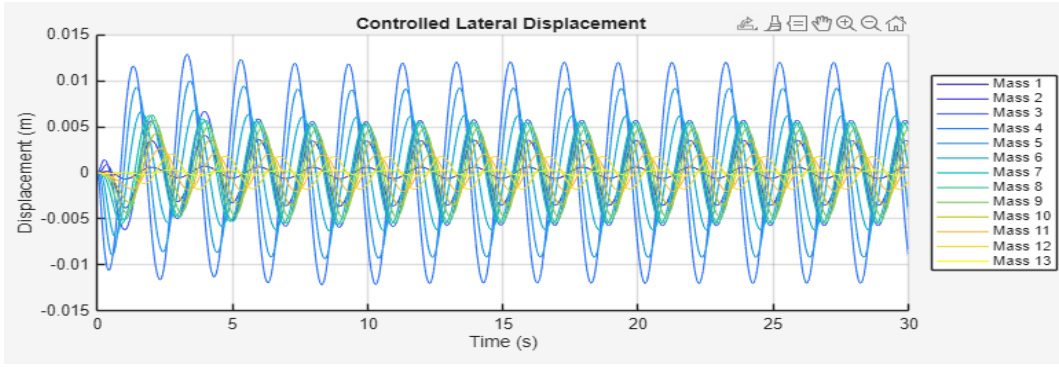


Figure 7: Controlled transverse displacement response

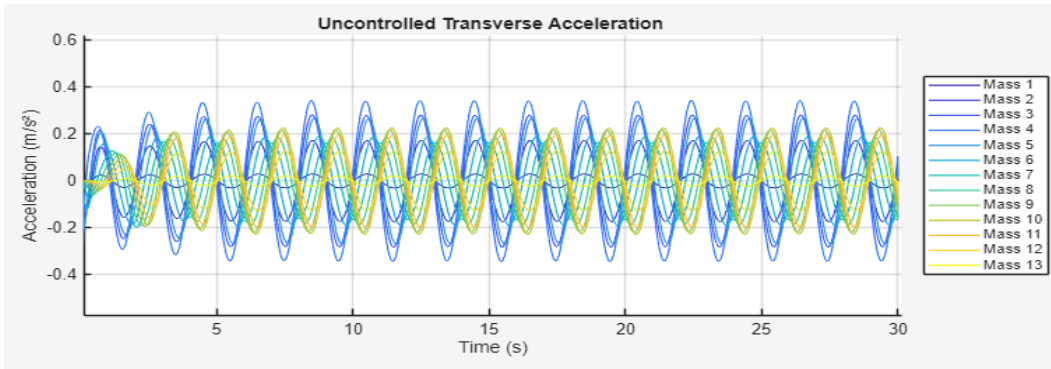


Figure 8: Uncontrolled transverse acceleration response to pedestrian input.

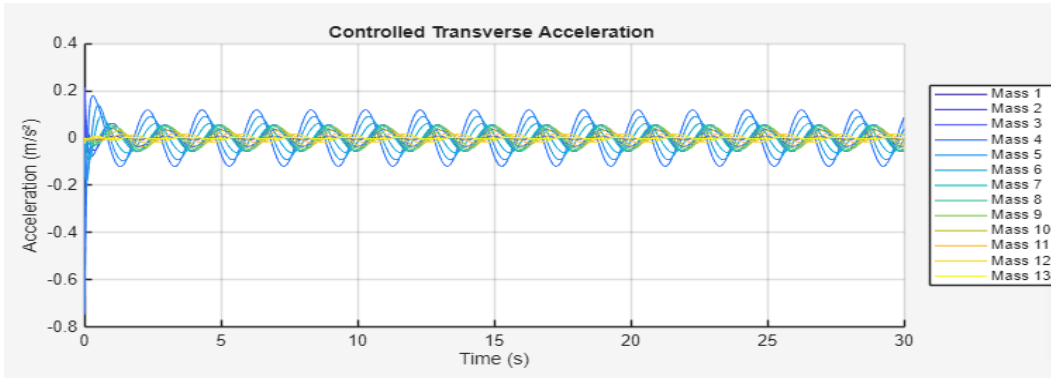


Figure 9: Controlled transverse acceleration response

The uncontrolled displacement response to a 39 pedestrian crowd at a synchronised step frequency of 0.5016 Hz for the discretised 13-point mass footbridge is shown in figure 6. The response shows a maximum displacement of 35 mm. The displacement response of the footbridge model under actuator control is shown in figure 7, showing a maximum displacement of 12 mm. This shows a 65.7 % improvement in the maximum displacement of the footbridge, showing the effectiveness of the controller. The acceleration response of the footbridge model without the control law is shown in figure 8, showing a maximum value of 0.338 ms^{-2} . The controlled response of the footbridge in figure 9, shows a maximum acceleration response of 0.166 ms^{-2} at the 0.4 second mark but settles at 0.114 ms^{-2} from the 1.17 second mark and onwards. This shows a 66.2 % reduction of the bridge's acceleration response.

3.2 3rd Lateral Vibration Mode

The response of the footbridge to the third lateral frequency with the calculated critical number of pedestrians is shown in the following figures.

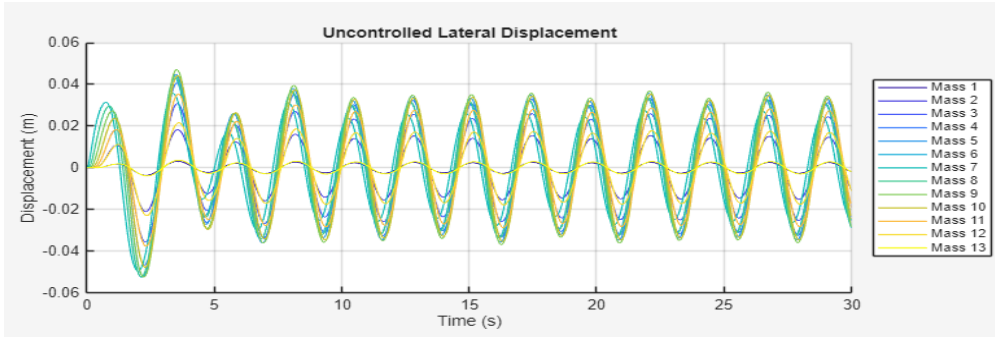


Figure 10: Uncontrolled transverse displacement response to pedestrian input

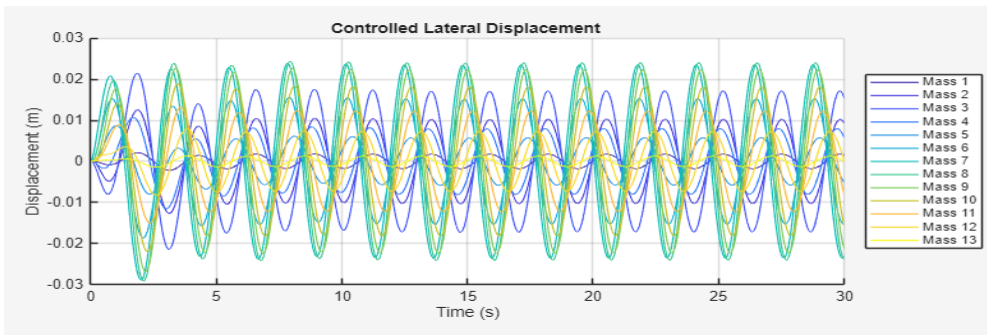


Figure 11: Controlled transverse displacement response

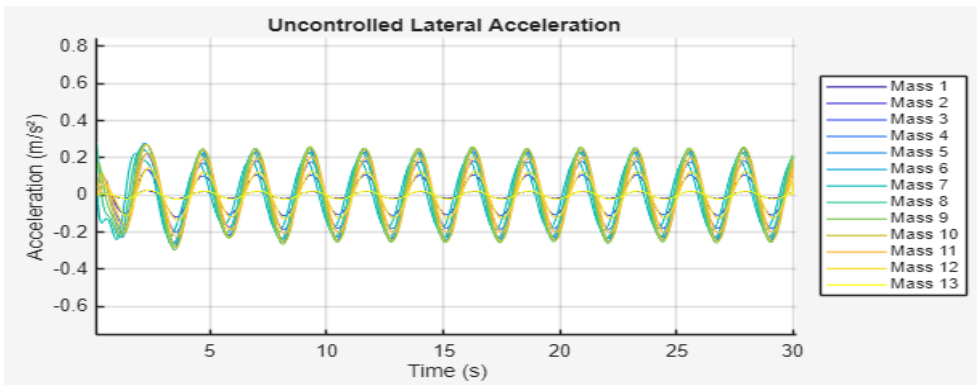


Figure 12: Uncontrolled transverse acceleration response to pedestrian input

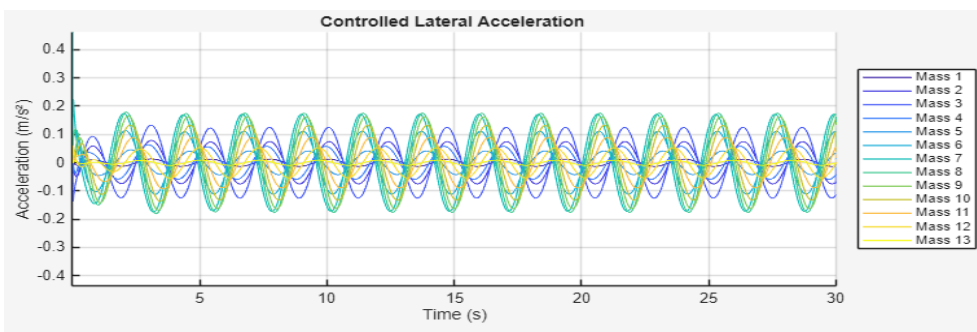


Figure 13: Controlled transverse acceleration response

The uncontrolled displacement response of the discretised 13-point mass footbridge subjected to an 86-person synchronised crowd at a step frequency of 0.74230 Hz is shown in Figure 10. The response reaches a maximum displacement of 45 mm at approximately 4.4 seconds. Under actuator control in figure 11 the maximum displacement reduces to 23 mm, representing a 48.8 % reduction. For acceleration, figure 12 presents the uncontrolled response of the

footbridge to a distributed lateral force, with a peak acceleration of 0.28 ms^{-2} . When control is applied in figure 13, the maximum acceleration reduces slightly to 0.18 ms^{-2} , an improvement of only 35.7%. The reduced effectiveness in controlling the third mode arises from a misalignment between actuator placement and the modal force distribution, which limits controllability and diminishes the actuators influence on this response.

4. CRITICAL SERVICEABILITY ANALYSIS

The criteria governing serviceability in footbridge design remains less clearly defined, particularly for lateral vibrations. While vertical and horizontal comfort limits are relatively well established in international standards, the guidance on lateral motion varies significantly and, in many cases, remains incomplete. There is no single regulatory body that has provided universally accepted thresholds for lateral accelerations or displacements making this a developing research area with room for interpretation. For this reason, the analysis adopted in this paper is a comparative approach referencing multiple international guidelines and codes to benchmark performance against a range of comfort and safety criteria. This ensures that the findings are not restricted to one geographical context, but can be applied more broadly. Footbridge serviceability in this context is governed primarily by pedestrian comfort, since excessive discomfort encourages gait adaptations and in group scenarios can lead to synchronisation effects that amplify bridge vibrations. Through alignment of the serviceability checks with the various thresholds specified across different standards, the analysis provides a robust basis for assessing the bridges performance under ATMD control.

The following table provides a summary of the footbridge performance for the 2nd to the 6th vibrational modes at the critical pedestrian number shown in table 7 below.

Table 7: Multimodal Footbridge Response

Lateral Mode	Uncontrolled Displacement	Controlled Displacement	Performance Improvement	Uncontrolled Acceleration	Controlled Acceleration	Performance Improvement
2	0.035	0.012	+ 65.7 %	0.338	0.114	+ 66.2 %
3	0.045	0.023	+ 48.8 %	0.28	0.18	+ 35.7 %
4	0.019	0.0102	+ 46.3 %	0.70	0.32	+ 54.2 %
5	0.022	0.00721	+ 67.2 %	1.103	0.36	+ 67.3 %
6	0.020	0.0055	+ 72.5 %	1.50	0.41	+ 72.6 %

The following table provides a summary of the serviceability limits for footbridges proposed by regulatory bodies.

Table 8: Serviceability limits

Code	Frequency	Acceleration (ms^{-2})
ISO 10137	1 – 1.7	< 0.21
Eurocode 5	0.5 – 2.5	< 0.70
British National Annex	< 1.5	0.5 – 2
SETRA [13]	0.2-2.5	Maximum comfort: < 0.15 Average comfort: 0.15 – 0.3 Minimum comfort: 0.3 – 0.8 Unacceptable: > 0.8
HIVOSS [14]	0.5 -1.2	Maximum comfort: < 0.1 Average comfort: 0.1 – 0.3 Minimum comfort: 0.3 – 0.8 Unacceptable: > 0.8

The results in table 7 demonstrate that the LQR controller delivers strong performance across all vibrational modes. For the second lateral mode, the controlled response meets all serviceability limits in table 8 achieving maximum comfort under the HIVOSS guidelines and average comfort under the SETRA guidelines. At the third lateral mode, the response also satisfies every serviceability requirement, with comfort levels rated as average in both SETRA and HIVOSS. In contrast, the fourth, fifth, and sixth lateral modes fall within the minimum comfort band for both SETRA and HIVOSS. Nevertheless, these responses remain below the Eurocode acceleration threshold. However, the stricter ISO standard is not satisfied, as the observed accelerations of 0.312 ms^{-2} , 0.36 ms^{-2} and 0.41 ms^{-2} exceed its 0.21 ms^{-2} limit. Overall, the findings confirm that the LQR controller substantially improves structural serviceability, with its effectiveness becoming especially pronounced at higher modes a notable strength of the approach.

4.1 Energy Consideration

The energy requirements of the proposed control law are provided in this section to grade feasibility of this system.

Table 9: Power consumption for each ATMD

Mode	Peak	Current	(A)	Peak	Voltage	(V)	Peak	Power	(W)
	1 st	2 nd	3 rd	1 st	2 nd	3 rd	1 st	2 nd	3 rd
2	4.90	3.55	4.40	25.14	9.33	10.05	91.23	32.54	42.16
3	1.91	8.94	4.50	8.08	23.95	10.96	9.95	193.69	43.75
4	0.74	1.39	3.65	2.76	3.85	10.62	1.41	4.18	21.41
5	2.19	12.82	1.35	6.79	34.27	3.27	12.67	439.40	3.56
6	1.49	4.61	2.68	5.78	12.92	10.55	7.26	58.89	27.79

The energy requirements of the proposed three ATMD system under LQR control show that the actuators experience considerable instantaneous power demand, particularly in the fifth mode where the combined peak power surpasses 400 W. Although such peaks are not sustained during operation, they raise important feasibility concerns. The large transient currents and voltages reported in table 9 could lead to amplifier saturation or current limiting, thereby reducing the effectiveness of the control law. Moreover, repeated short duration peaks may accumulate thermal stress in the actuator coils and associated power electronics, potentially causing overheating if not managed appropriately. From a design perspective, these findings suggest that the power supply and amplifiers must be specified with adequate safety margins typically 1.5 to 2 times the observed maximums of 34.27 V and 12.82 A to ensure reliable operation. Practical implementation can draw on established thermal management strategies such as the use of heat sinks and thermally conductive mounts for the actuator coils safeguarding performance. Transient peaks can be mitigated through controller level strategies such as rate limiting or the installation of fast acting current limiters while maintaining efficiency and long-term reliability. Overall, the analysis confirms that the system remains feasible provided that a stable continuous power supply is available as the suppression capability of the passive components is not enough to stabilise the footbridge. The controller design should also accommodate transient power surges while maintaining efficiency and long-term reliability.

4.2 Design evaluation

The table of footbridge responses demonstrate that the system meets some of the specified performance targets across the excited modes of vibration. In regards to the specification for lateral excitation performance using the better defined SETRA and HIVOSS guidelines, the control system manages to meet design specifications across all modes of vibration with all responses below the 0.80 ms^{-2} limit. This ensures that the stability specification is also met across all vibrational modes. The power consumption values provided in table 9, show that across all modes of excitation the maximum voltage requirement of $\pm 50\text{V}$ is met with the maximum achieved voltage at $\pm 34.27 \text{ V}$ in the fifth mode. Current demand peaks at 12.82 A in the fifth mode, slightly exceeding the 10 A nominal rating, but still within feasible limits if proper headroom and thermal management are provided. Peak instantaneous power reaches 439.40 W remaining below the 500 W specification, ensuring adequate provision for transient surges. These results confirm that the system can be reliably powered without the risk of amplifier saturation, provided supply and cooling are appropriately provisioned. In summary the control system achieves substantial vibration suppression and complies with power and voltage requirements, failing only the current specification by a 2.82 A difference. As a result, satisfaction of the stability and lateral response specifications in table 5 adds confidence into the implementation of the proposed control law.

5. SOCIAL, ECONOMIC & ENVIRONMENTAL IMPACT

5.1 Social Impact

The adoption of the proposed control system has clear social benefits by improving pedestrian comfort and safety on the footbridge. Lower vibration levels reduce the likelihood of gait adaptations and synchronisation effect, which are known to heighten discomfort and create unsafe conditions in crowds. By enhancing user confidence, the bridge becomes more accessible to vulnerable groups such as the elderly and people with mobility impairments. At the same time social risks exist with the visibility of mechanical devices or concerns over system failure affecting public perception of safety. To mitigate this, clear communication about the purpose and reliability of the system is important. There is also an equality consideration as advanced control solutions are more likely to be implemented on high profile bridges and not within smaller communities putting those people at risk. To counter that the proposed control system should be implemented on any available bridge structures that experience lateral vibrations. Overall, when properly integrated the control system

can significantly improve the social value of pedestrian bridges by ensuring comfort, inclusivity and confidence in their use.

5.2 Economic Impact

From an economic perspective implementing ATMDs provides an alternative to costly structural modifications by increasing bridge lateral stiffness post construction [15]. Through lateral vibration mitigation, the system helps avoid revenue losses associated with bridge closures or reduced capacity during public events, while also extending the service life of the structure. Although the initial investment in hardware, installation and commissioning is significant as shown by the Crystal Tower vibration mitigation system which cost approximately \$ 350 000 [16], this is often outweighed by the long-term savings compared with extensive retrofitting or reconstruction. The low energy consumption for continuous operation shown in the design evaluation, ensures that the proposed solution remains an economically viable solution throughout its operation. In balance, the ATMD system offers a cost effective and adaptable solution particularly where bridges are of strategic, cultural or commercial importance.

5.3 Environmental Impact

Environmentally, ATMDs represent a relatively low impact solution compared with large scale structural interventions. By enabling retrofitting rather than full reconstruction, they help preserve the embodied carbon in existing infrastructure and avoid the environmental costs associated with producing, transporting, and installing new structural materials. Operational energy use is generally modest ensuring a relatively low carbon footprint particularly outside periods of peak demand, further enhancing sustainability. Potential environmental drawbacks include noise emissions from actuators and end-of-life disposal of components; these can be mitigated through careful specification, maintenance, and recycling strategies. Beyond direct environmental impacts, improving pedestrian comfort also promotes walking as a sustainable mode of transport. Overall, when designed and managed with lifecycle considerations in mind, ATMDs offer meaningful environmental benefits while supporting sustainable urban mobility.

6. SUSTAINABILITY ANALYSIS

The long-term viability of the ATMD based control system depends on its ability to operate reliably under South African environmental and infrastructural conditions. In particular the reoccurring issue of load shedding and grid instability poses a major challenge to maintaining continuous operation for active control. To address this, the system must incorporate renewable energy backup such as a solar photovoltaic (PV) system combined with battery storage ensuring that the ATMDs remain functional during power outages, this approach would enhance the resilience of the solution. The power consumption analysis provided in table 9, shows that the peak instantaneous power demand is 439 W, with all actuators individually remaining below the 500W design limit. This relatively low power demand makes the integration of a solar battery system feasible as the daily energy requirement for the three ATMDs is low compared to typical small scale solar generation. Durability is also a key sustainability factor, thus the ATMDs and associated sensors must be designed with weather resistant enclosures to adapt to the -20°C to 40°C [17] temperature range in South Africa. Corrosion protection material and high-quality components must be used to ensure a minimum design life of 20 years with limited maintenance. This reduces resource consumption over the lifecycle of the system and minimises operational costs for the municipality or operator. In evaluating sustainability, the solution demonstrates several strengths as it significantly enhances structural safety and serviceability contributing to the longevity of the bridge by reducing structure fatigue during pedestrian loading and can be powered sustainably through solar power. The implementation of renewable backup directly ensures that the solution is technically, socially and environmentally sustainable for the South African environment.

7. CONCLUSION

This study has addressed the problem of pedestrian-induced lateral vibrations in footbridges through the design and evaluation of an active controller. Using MATLAB and Simulink, three control strategies were developed, with the first two validated on a simplified SDOF model as proof of concept, and the final LQR-based controller implemented on a more realistic representation. The results confirm that the proposed LQR control law can substantially reduce all lateral vibration modes, bringing responses within accepted serviceability limits. From a feasibility perspective, the energy demand of the controller, suggests that implementation is practical within certain operational bounds. This underlines the effectiveness of active control to enhance pedestrian comfort and structural safety in modern footbridge design. Future work could explore more adaptive control approaches such as intelligent controllers to extend the robustness and applicability of the proposed framework.

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Appendix A: Detailed Graduate Attribute mapping

Table A: Graduate Attributes Summary

GA	Section addressed	GA	Section addressed	GA	Section addressed
1	1, 2 and 3	2	1,2 and 3	3	2,3 and 4
5	Throughout the report	6a	Throughout the report	7a	5
7b	6	8	Throughout the report		

Appendix B: Visualisation of vibrations modes

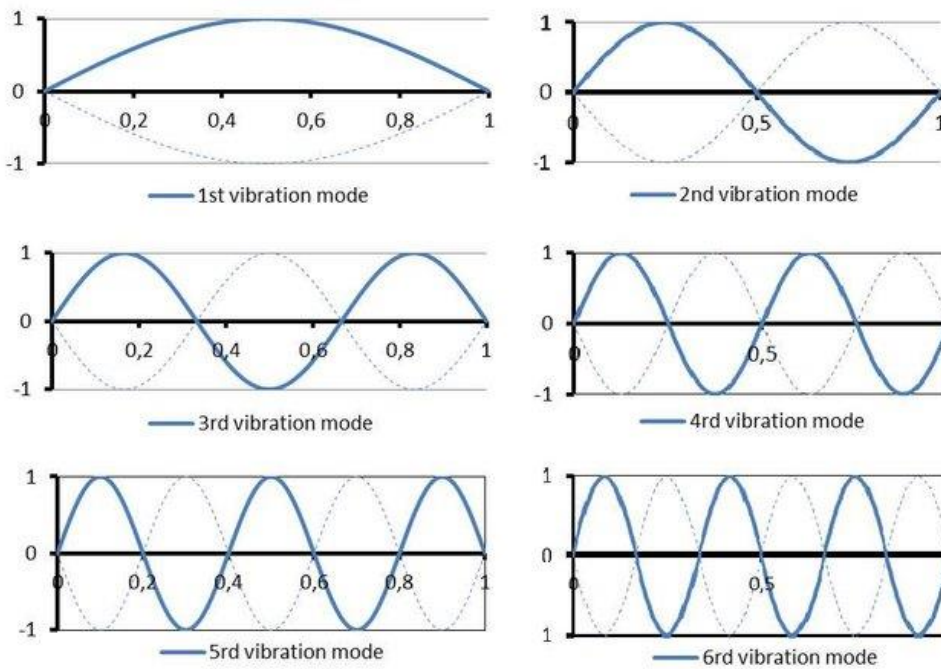


Figure B; Visualisation of different vibration modes for the footbridge model [18].

Appendix C: First lateral vibration mode Responses

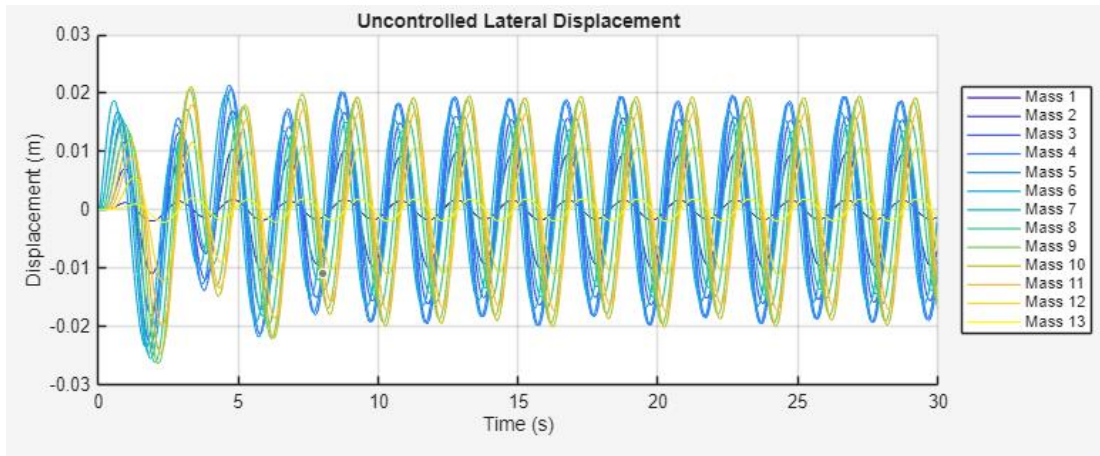


Figure C1: Uncontrol lateral displacement for the first vibrational mode

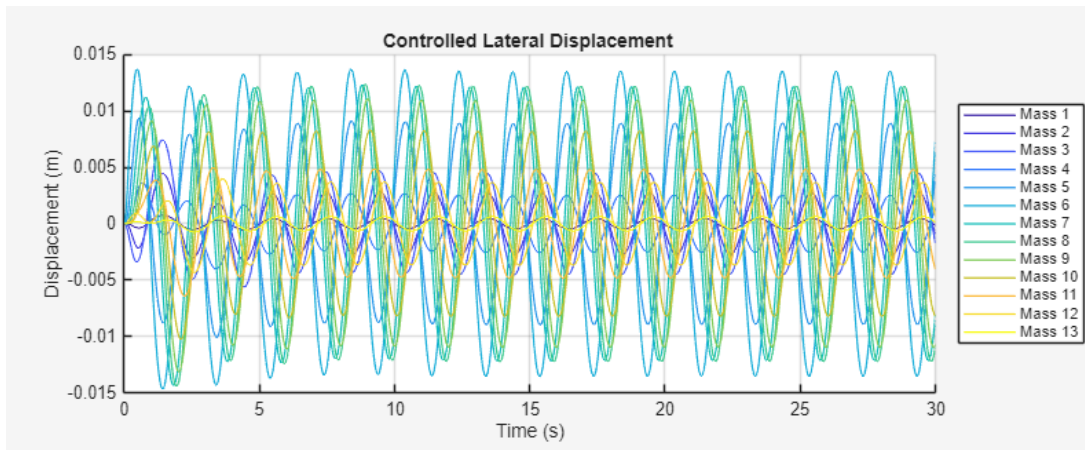


Figure C2: Controlled lateral displacement for the first vibrational mode

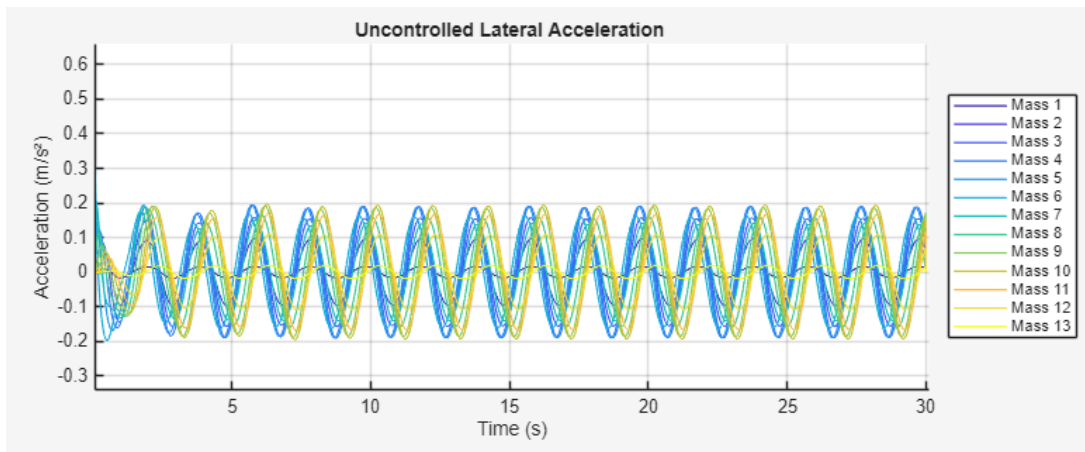


Figure C3: Uncontrolled lateral acceleration for the first mode of vibration

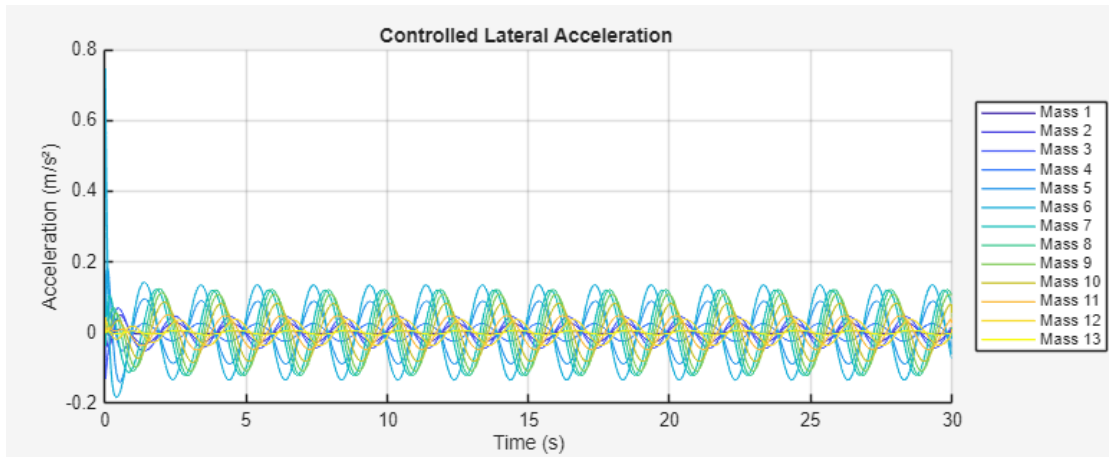


Figure C4: Controlled lateral acceleration for first mode of vibration

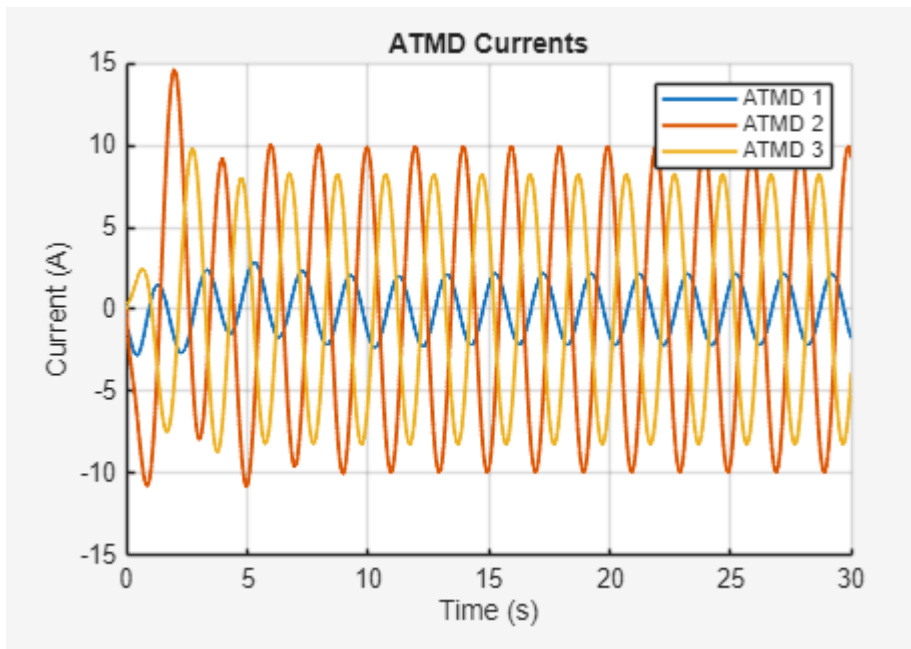


Figure C5: Individual current demands for the first mode of vibration

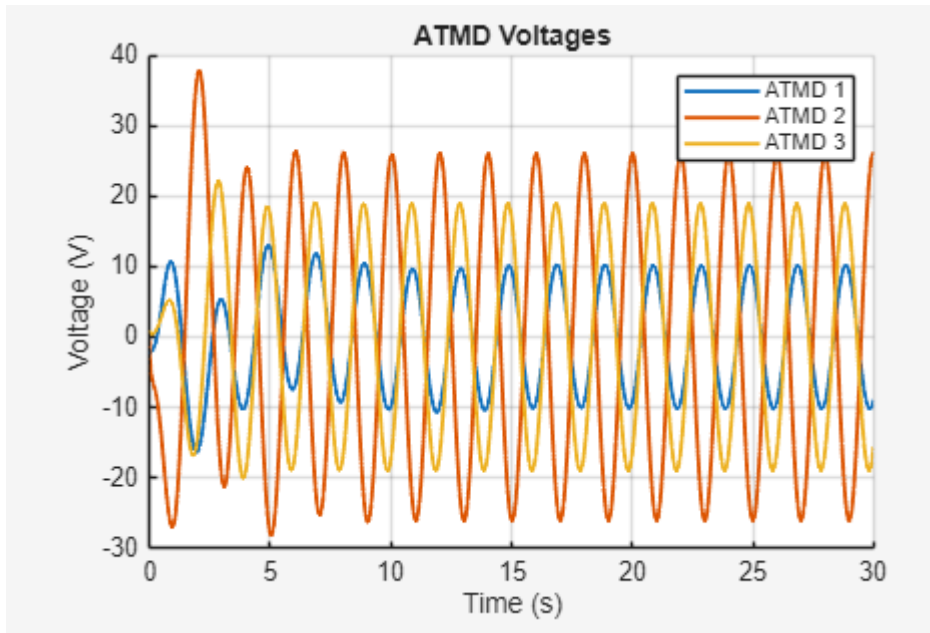


Figure C6: Individual voltage demands for the first mode of vibration

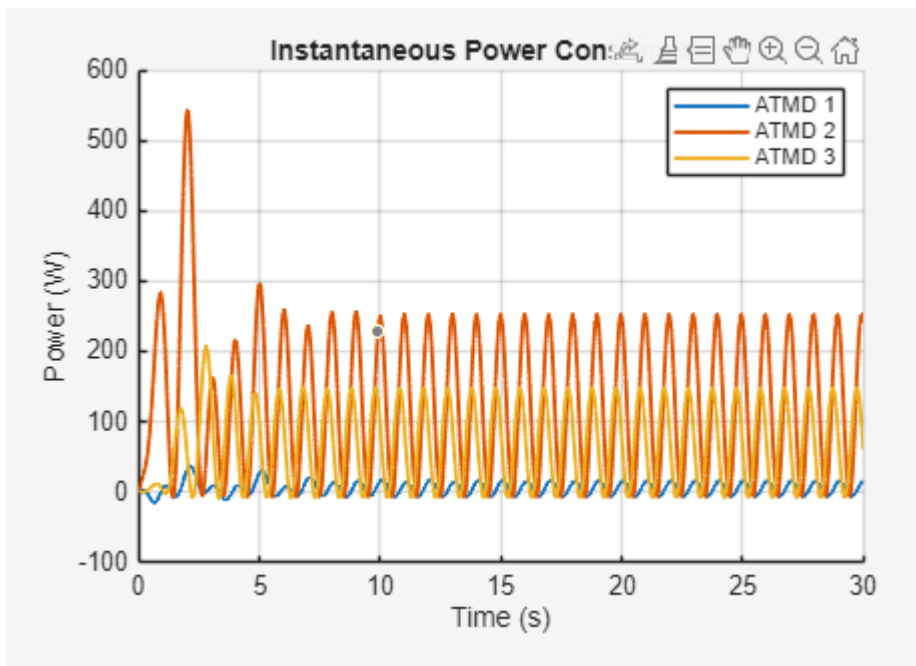


Figure C7: Individual power demands for the first mode of vibration

Appendix D: Closed loop poles

LQR controller designed for transverse motion. Closed-loop poles (first 10):

$-0.2573 \pm 0.5370j$ ($f=0.085$ Hz, $\zeta=0.432$)
 $-0.2573 \pm -0.5370j$ ($f=0.085$ Hz, $\zeta=0.432$)
 $-0.6105 \pm 1.3410j$ ($f=0.213$ Hz, $\zeta=0.414$)
 $-0.6105 \pm -1.3410j$ ($f=0.213$ Hz, $\zeta=0.414$)
 -0.7023
 $-0.9528 \pm 2.1253j$ ($f=0.338$ Hz, $\zeta=0.409$)
 $-0.9528 \pm -2.1253j$ ($f=0.338$ Hz, $\zeta=0.409$)
 -1.9684
 $-2.3802 \pm 4.1368j$ ($f=0.658$ Hz, $\zeta=0.499$)
 $-2.3802 \pm -4.1368j$ ($f=0.658$ Hz, $\zeta=0.499$)

Figure D: Results from closed loop analysis

Appendix E: Simulink SDOF model

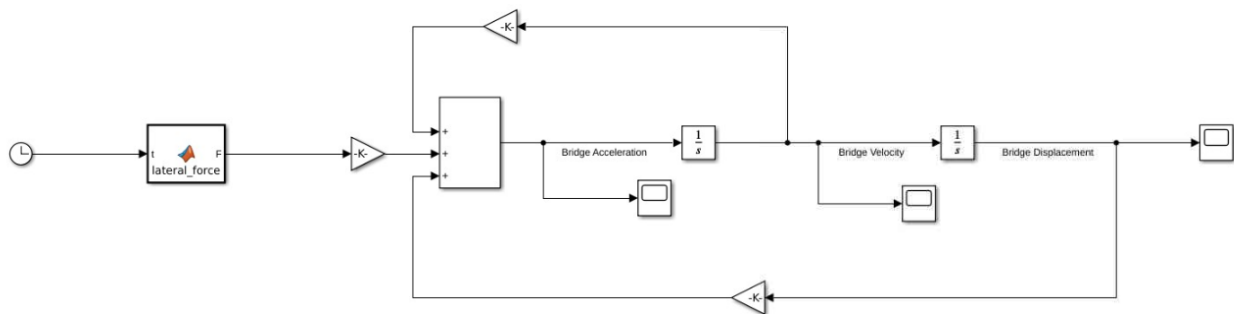


Figure E: Simulink SDOF footbridge model

Appendix F: PID controller layout in Simulink

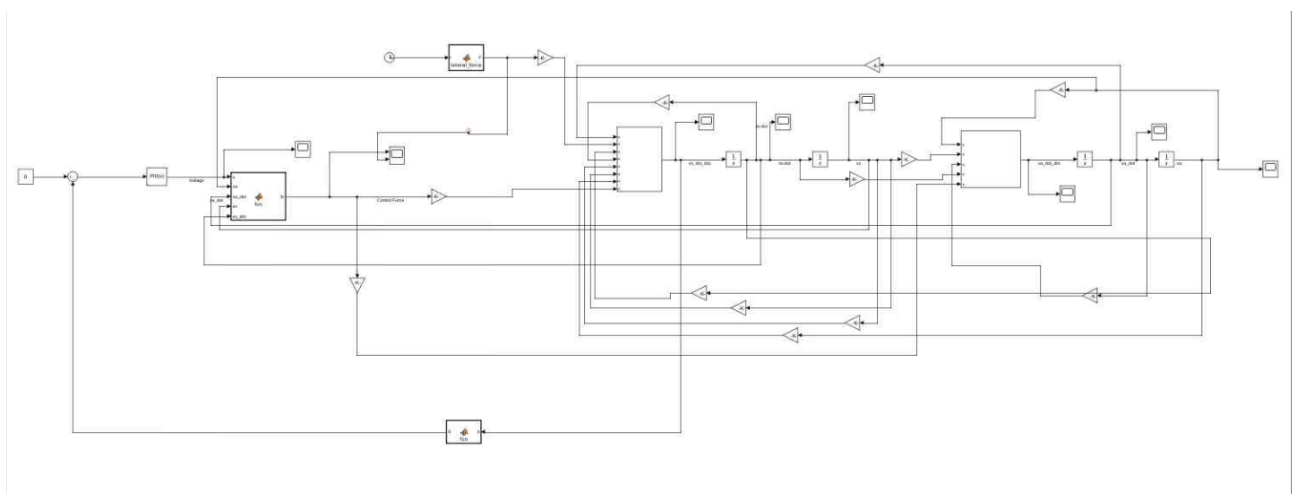


Figure F: PID controller simulation layout in Simulink

Appendix G: LQR simulation of the SDOF model

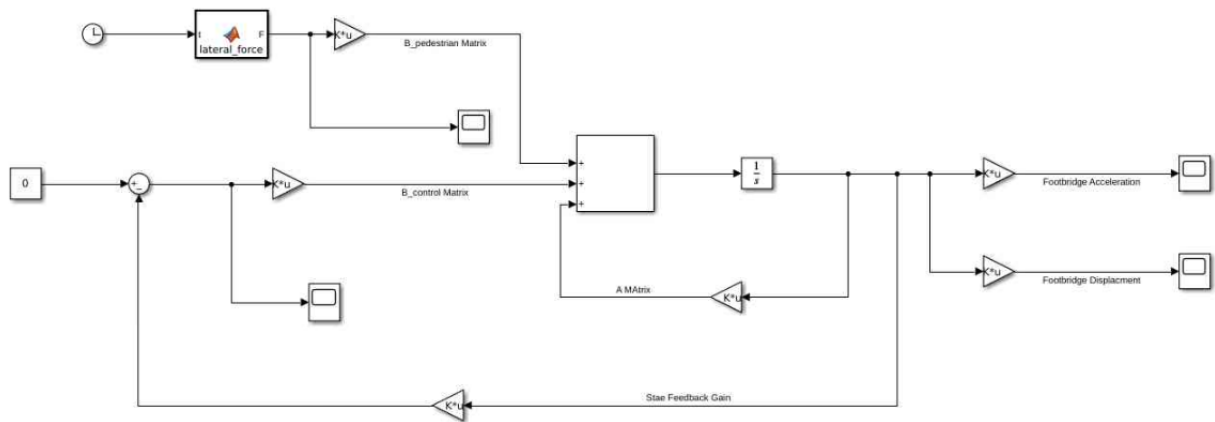


Figure G: Simulink blocks for the LQR simulation of the SDOF footbridge model